

ASSET DEFENDER - PROFILE + MESSAGE FORMULAS

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PROFILE FORMULAS (SET 1 - 10 CORE SIGNALS)

1) Follower-Following Ratio (FFR)

Formula:

$$\text{FFR} = \text{Followers} / \text{Following}$$

Interpretation:

- $\text{FFR} < 0.1$ -> Highly suspicious
- 0.1 to 0.5 -> Possible spam
- 0.5 to 2 -> Normal user
- > 5 -> Influencer/brand pattern

2) Engagement Rate (ER)

Formula:

$$\text{ER} = (\text{Likes} + \text{Comments}) / \text{Followers} * 100$$

Typical ranges:

- Normal user: 3% to 10%
- Influencer: 2% to 6%
- Fake influencer: $< 1\%$
- Brand: 1% to 3%

3) Post Activity Score (PAS)

Formula:

$$\text{PAS} = \text{Total Posts} / \text{Account Age (months)}$$

Interpretation:

- 0 -> Empty account
- 0.1 to 1 -> Inactive
- 1 to 10 -> Normal
- > 20 -> Spam/Bot tendency

4) Profile Completeness Score (PCS)

Formula:

$$\text{PCS} = (\text{Bio} + \text{ProfilePicture} + \text{Posts} + \text{Highlights} + \text{ExternalLink}) / 5$$

Binary feature coding:

- Present = 1
- Absent = 0

Interpretation:

- 0 to 0.2 -> Fake/empty
- 0.2 to 0.5 -> Suspicious
- 0.5 to 0.8 -> Normal
- ≥ 0.8 -> Strong profile

5) Username Similarity Score (Impersonation)

Formula:

$$\text{Similarity} = 1 - (\text{EditDistance} / \text{MaxLength})$$

Method:

- Use Levenshtein distance

Rule:

- $\text{Similarity} > 0.7$ -> Possible impersonation

6) Scam Keyword Score (SCS)

Formula:

$$\text{SCS} = \text{ScamKeywords} / \text{TotalWords}$$

Common keywords:

earn, profit, guaranteed, investment, crypto, forex, income, DM

7) External Link Risk Score (ELS)

Formula:

$$\text{ELS} = \text{RiskyLinks} / \text{TotalLinks}$$

Risky domains examples:

bit.ly, tinyurl, telegram, wa.me, t.me

8) Comment Authenticity Score (CAS)

Formula:

$$\text{CAS} = \text{UniqueComments} / \text{TotalComments}$$

Low CAS indicates copied/bot-style engagement.

9) Content Consistency Score (CCS)

Formula:

$$\text{CCS} = \text{SimilarTopicPosts} / \text{TotalPosts}$$

Large sudden topic shifts can indicate compromise/hijack.

10) Overall Risk Score (Set 1)

Formula:

RiskScore =

$$\begin{aligned} & (0.20 * \text{FFR_score}) + \\ & (0.20 * \text{Engagement_score}) + \\ & (0.15 * \text{PCS_risk}) + \\ & (0.15 * \text{Username_similarity_risk}) + \\ & (0.15 * \text{Scam_keyword_score}) + \\ & (0.15 * \text{External_link_score}) \end{aligned}$$

Interpretation:

- 0.0 to 0.3 -> Legit
- 0.3 to 0.6 -> Suspicious
- 0.6 to 1.0 -> High risk

PROFILE FORMULAS (SET 2 - 30 SIGNALS, 6 LAYERS)

LAYER 1 - Profile Identity Signals

1. Username Similarity:

$$\text{Similarity} = 1 - (\text{EditDistance} / \text{MaxLength})$$

2. Username Complexity Score:

$$\text{UCS} = (\text{numbers} + \text{special_characters}) / \text{username_length}$$

3. Profile Completeness:

$$\text{PCS} = (\text{bio} + \text{profile_photo} + \text{posts} + \text{highlights} + \text{link}) / 5$$

4. Bio Keyword Risk:

$$\text{BKR} = \text{scam_keywords} / \text{total_words}$$

5. External Link Risk:

$$\text{ELR} = \text{risky_links} / \text{total_links}$$

LAYER 2 - Network Signals

6. Follower-Following Ratio:

$$\text{FFR} = \text{followers} / \text{following}$$

7. Follower Growth Spike:

$$\text{GrowthRate} = \text{followers_today} - \text{followers_yesterday}$$

8. Mutual Follower Density:

$$\text{MFD} = \text{mutual_followers} / \text{total_followers}$$

9. Bot Follower Ratio:

$$\text{BFR} = \text{suspected_bot_followers} / \text{total_followers}$$

10. Follower Quality Score:

$$\text{FQS} = \text{real_followers} / \text{total_followers}$$

LAYER 3 - Content Signals

11. Post Frequency:

$$PF = \text{posts} / \text{account_age_months}$$

12. Content Consistency:

$$CC = \text{dominant_topic_posts} / \text{total_posts}$$

13. Caption Spam Score:

$$CSS = \text{scam_keywords} / \text{caption_words}$$

14. Hashtag Spam Score:

$$HSS = \text{hashtags} / \text{caption_words}$$

15. Duplicate Content Score:

$$DCS = \text{repeated_posts} / \text{total_posts}$$

LAYER 4 - Engagement Signals

16. Engagement Rate:

$$ER = (\text{likes} + \text{comments}) / \text{followers} * 100$$

17. Comment Authenticity:

$$CAS = \text{unique_comments} / \text{total_comments}$$

18. Like Velocity:

$$LV = \text{likes} / \text{first_hour}$$

19. Comment Spam Ratio:

$$CSR = \text{spam_comments} / \text{total_comments}$$

20. Engagement Consistency:

$$EC = \text{stddev}(\text{likes_per_post})$$

LAYER 5 - Behavioral Signals

21. Follow Rate:

$$FR = \text{follows_per_day}$$

22. Unfollow Rate:

$$UR = \text{unfollows_per_day}$$

23. DM Spam Score:

$$DSS = \text{scam_messages} / \text{total_messages}$$

24. Activity Time Pattern:

$$ATP = \text{activity_hours_variance}$$

25. Posting Automation Score:

$$PAS = \text{posts_same_interval} / \text{total_posts}$$

LAYER 6 - Risk Context Signals

26. Account Age:

$$AA = \text{current_date} - \text{creation_date}$$

27. Username Change Frequency:

$$UCF = \text{username_changes} / \text{months}$$

28. Bio Change Frequency:

$$BCF = \text{bio_changes} / \text{months}$$

29. Report Frequency:

$$RF = \text{reports} / \text{followers}$$

30. External Platform Redirection:

$$EPR = \text{external_redirect_links} / \text{posts}$$

Final Risk Score (Set 2):

RiskScore =

$$0.15 * \text{ProfileSignals} +$$

$$0.20 * \text{NetworkSignals} +$$

$$0.20 * \text{ContentSignals} +$$

$$0.20 * \text{EngagementSignals} +$$

$$0.15 * \text{BehavioralSignals} +$$

$$0.10 * \text{ContextSignals}$$

Risk Classification:

- 0.0 to 0.3 -> Legitimate
- 0.3 to 0.5 -> Low risk
- 0.5 to 0.7 -> Suspicious
- 0.7 to 1.0 -> High risk

MESSAGE FORMULAS (CURRENT PROJECT)

MESSAGE RISK SCORE (0-100)

Raw formula:

$R_{raw} = 6 + C_{cred} + C_{transfer} + C_{code} + C_{imp} + C_{press} + C_{fin} + C_{link} + C_{switch} + C_{bot} + B_{conv} - S_{context}$
 $R = \text{clamp}(\text{round}(R_{raw}), 0, 100)$

Components:

- $C_{cred} = \min(35, \text{credentialHits} * 10)$
- $C_{transfer} = \min(30, \text{credentialTransferHits} * 12)$
- $C_{code} = \min(18, \text{genericCodeRequestHits} * 8)$
- $C_{imp} = \min(18, \text{impersonationHits} * 6)$
- $C_{press} = \min(18, \text{pressureHits} * 6)$
- $C_{fin} = \min(16, \text{cryptoContextHits} * 4)$
- $C_{link} = \min(16, \text{links} * 4) + \min(20, \text{suspiciousLinks} * 10) + \min(16, \text{riskyTldHits} * 8)$
- $C_{switch} = (\text{platformSwitchHits} > 0 \text{ and } \text{redirectionIntentHits} > 0 ? 12 : 0) + \min(8, \text{platformSwitchHits} * 2)$
- $C_{bot} = (\text{repetitionRatio} \geq 0.45 ? 14 : 0)$
- $B_{conv} = \min(16, \text{conversationRiskSignalCount} * 4)$
- $S_{context} = \text{clamp}(\text{clamp}(\text{clamp}(\text{safetyNegationHits} > 0 ? \min(18, \text{safetyNegationHits} * 10) : 0) + (\text{reportingContextHits} > 0 \text{ and } \text{credentialTransferHits} == 0 \text{ and } \text{suspiciousLinks} == 0 ? \min(12, \text{reportingContextHits} * 6) : 0), 0, 24)$

Hard-risk floors:

- OTP + crypto context present and $R < 60 \Rightarrow R = 60$
- explicit credential transfer + impersonation/pressure/crypto and $R < 70 \Rightarrow R = 70$
- links + credential request and $R < 70 \Rightarrow R = 70$

Class mapping from message risk score:

- $R \geq 70 \rightarrow$ high-risk
- $40 \leq R < 70 \rightarrow$ suspicious
- $R < 40 \rightarrow$ likely-human

MESSAGE CLASS SCORES (bot/spam/scam/phishing/hacker/mixed)

Per-class formulas:

- $S_{phishing} = \text{clamp}(24 * \text{phishingLinkCount} + 8 * \text{credentialKeywordCount} + 8 * \text{impersonationKeywordCount} + 10 * \text{brandTypoquatCount} + 8 * \text{suspiciousDomainCount} + 6 * \text{riskyTldCount} + 5 * \text{obfuscatedLinkSignals} + 4 * \text{suspiciousPathHits})$
- $S_{hacker} = \text{clamp}(22 * \text{credentialTransferCount} + 10 * \text{genericCodeRequestCount} + 7 * \text{credentialKeywordCount} + 7 * \text{impersonationKeywordCount} + 6 * \text{financialPressureCount} + 10 * \text{otpCryptoCombo} + 8 * (\text{riskScore} \geq 65))$
- $S_{scam} = \text{clamp}(9 * \text{scamKeywordCount} + 10 * \text{financialPressureCount} + 6 * \text{cryptoContextCount} + 10 * \text{otpCryptoCombo} + 15 * \text{scamCount} + 6 * \text{broadcastHits} + 10 * (\text{platformSwitch and redirectionIntent}) +$

0.25*riskScore)

- S_spam = clamp(40*repetitionRatio + 8*consecutiveRepeatCount + 8*broadcastHits + 1.2*suspiciousKeywordCount + 6*(totalMessages>=25))
- S_bot = clamp(45*repetitionRatio + 10*consecutiveRepeatCount + 40*rapidFireRatio + 6*maxBurst2Min + 20*shortMessageRatio + 8*(incomingRatio>0.85))

Mixed-risk arbitration:

Let:

- top = max(S_i)
- second = second_max(S_i)
- margin = top - second
- highClassCount = count(S_i >= 45)

Emit mixed-risk when:

- highClassCount >= 2 and margin <= 8, or
 - highClassCount >= 3
- with top >= 45.

Otherwise:

- if top >= 40, emit top class (phishing-risk/hacker-risk/scam/spam/bot)
- else emit likely-human (or suspicious-message if global risk remains elevated)